

Ajay Lohith Kulla

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Career Objective

Final-year B.Tech graduate with hands-on experience in full-stack development, backend services, and RESTful API integrations. Proficient in Java, Spring Boot, and React with exposure to relational databases, version control, and secure coding practices. Eager to contribute to enterprise application development, system integrations, and modernization initiatives in a collaborative Agile environment..

Skills

- **Languages:** Java, JavaScript, C/C++
- **Backend:** Spring Boot, Spring MVC, Hibernate, Kafka, Redis, RESTful APIs
- **Frontend:** React.js, HTML, CSS, Javascript
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Tools:** Git, GitHub, Docker, Postman, Firebase, IntelliJ, Groq LLM, VS code

Projects

Distributed Key-Value Store – Fault-Tolerant Distributed System GitHub

- **Architected** a distributed key-value store using **Spring Boot microservices** and **API Gateway**, implementing load balancing across 3 independent nodes for high availability.
- **Engineered** event-driven data replication using **Kafka** and a Read/Write-Through caching layer with **Redis**, ensuring fault tolerance and **99.8%** replication consistency.
- **Implemented** full CRUD operations with **PostgreSQL** persistence, idempotent Kafka event consumption, and optimistic locking to handle concurrent writes and prevent data conflicts.
- **Deployed** all services as isolated containers using **Docker Compose** with dedicated Kafka topics for create, update, delete, and node-status events enabling seamless cluster synchronization.
- **Tech Stack:** Java, Spring Boot, Microservices, Kafka, Zookeeper, Redis, PostgreSQL, Docker, React

SnapLink – High-Performance URL Shortener Live | GitHub

- **Architected** a redirect engine using **Spring Boot** and **Redis**, achieving **<10ms** latency via a **Cache-Aside pattern** to optimize high-traffic URL resolution.
- **Implemented Base62 encoding** with **SecureRandom** to support 3.5T+ permutations, utilizing an automated retry logic to resolve **hash collisions** through DB integrity constraints.
- **Engineered** an **IP-based Fixed Window rate limiter** in Redis and custom alias **Regex validation**, ensuring **99.9% system uptime** and protection against API abuse.
- **Designed** a **Fail-Open** architecture to maintain service availability during Redis outages by falling back to indexed **PostgreSQL** lookups and **HTTP 302 redirects**.
- **Tech Stack:** Java, Spring Boot, Redis, PostgreSQL, Docker, RESTful APIs

PlacementBoard - Career Development Platform Live | GitHub

- **Developed** a full-stack placement platform with modules for jobs, companies, interviews, and articles, improving user engagement by **35%** through centralized access to resources.
- **Implemented JWT-based authentication** with **RBAC** and OTP-based password reset, reducing unauthorized access risks by **25%** and improving account recovery efficiency.
- **Integrated Groq (LLaMA 3.3)** for AI-powered job postings and article generation, increasing content availability and reducing manual effort by **40%**.
- **Tech Stack:** Spring Boot, Java, React, JavaScript, MongoDB, Vite, Docker, Groq AI, JWT, Tailwind CSS

Internship

Zidio Web Development (Remote) Jun 2024 – Jul 2024

- Contributed to full-stack web projects using the MERN stack.
- Built dynamic React components and integrated RESTful APIs.

INDO-GERMAN Institute of Advanced Technology May 2024 – Jun 2024

- Analyzed microcontroller workflows and documented technical processes.